Erfan **Khadem**

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An Undergraduate EE Student Looking for Research Opportunities

Education



Bachelors of Science in Electrical Engineering — *2022 - 2026*
Sharif University of Technolog — *Tehran, Iran*

- Optional Courses: DSP | Compressed Sensing | Operating Systems Lab | Networks Lab | Cryptography | Graphical Models
- GPA: 18.60 / 20, Top 12%
- GPA of EE Courses: 19.15 / 20, Top 8%



Highschool Diploma in Physics and Mathematics — *2019 - 2022*
Dastgheib 1 Highschool of Shiraz — *Shiraz, Iran*

- GPA: 19.75 / 20

Professional Experience



Embedded Hardware and Backend Developer — *2023*
Home Automation Project at National Elites Foundation — *Tehran, Iran*

- Worked alongside a team of computer security experts, electrical engineers and developers to build home automation appliances and smart gateways using TI's Zigbee chips and Espressif System's ESP32 and ESP8266 microcontroller series
- Technologies:
  - FreeRTOS
  - Zigbee
  - Websocket
  - Embedded Linux



Fullstack Developer and Systems Administrator — *2022 - 2023*
Abarkelas — *Tehran, Iran*

- Fix bugs and improve UX at our frontend using SSR and Vue.js
- Improve monitoring and alerting for our backend systems
- Implement self-healing procedures for our backend systems
- A/B test new features to guide our product development process
- Technologies:
  - Django REST Framework
  - Vue.js
  - Docker
  - Postgres + Redis
  - Grafana + Prometheus



Fullstack Developer and Systems Administrator — *2020 - 2022*
Niroutrans Co. — *Shiraz, Iran*

- Development of an advanced monitoring system for Niroutrans' solar power plant, the biggest roof-mounted solar power plant in Iran
- Design and implement online product documentation center
- Technologies:
  - Microsoft SQL Server + Redis
  - Bootstrap 4 & 5
  - Modbus TCP
  - Protobuf
  - Nginx

Projects & Associations

RP2040 based Intel 8080 emulator
2024-2026

erfan-khadem/rpi8080 — *Github*

- Implemented a full-fledged Intel 8080 emulator on the Raspberry Pi Foundation's RP2040.
- The project has been used since Spring 2024 in the computer architecture laboratory for educational purposes.
- On-chip debugger, virtual input and output ports and accompanying PC software for ease of use.

Breathing Rate Detection Using WiFi Channel State Information
2024

A Hobby Project

- Breathing rate detection using a common off the shelf intel network card.
- Achieves an accuracy of more than 98 % even in highly congested areas.
- Runs transparently and does not disrupt the normal user activity.
- Uses ICA for Blind Source Separation

Linear OptoCoupler
2024

Electronics Lab Project

- A highly linear, optically isolated analog voltage isolator using `HCNR201`.
- Achieves a maximum signal bandwidth of more than $1\ \mathrm{MHz}$.
- Less than $-70\ \mathrm{dB}$ THD at $1\ \mathrm{kHz}$ and full swing.
- PCB was designed using KiCAD.

Advanced Algorithms in Rust
2022

TheAlgorithms/Rust — *Github*

- Implemented advanced graph algorithms such as centroid decomposition and max-flow.
- Implemented various cryptographic algorithms including Salsa20 and HMAC.
- Improved and optimized the implementation of various algorithms like SHA256 and DSU.
- Many other big and small improvements and implementation.

SSH Anti-Censorship Tool
2023

SSH-liberty-bridge — *Github*

- Implemented in go with the user management component written in python.
- Has been used successfully for bypassing GFW restrictions in Iran and bringing internet freedom to the citizens.
- Integrated in Hiddify's excellent GFW bypass solution.

Certificates

2021 **Scientific Computing with Python**, Free Code Camp *view*
2021 **Javascript Algorithms and Data Structures**, Free Code Camp *view*

Skills

Languages English | Persian
Software Stack C/C++ (Embedded/FreeRTOS/Linux) | Python (Pandas/Numpy/Scipy/Django) | JS (Svelte/Vue.js) | Rust | Go
Hobbies Chess | Bicycle | Electronics | Mathematics